

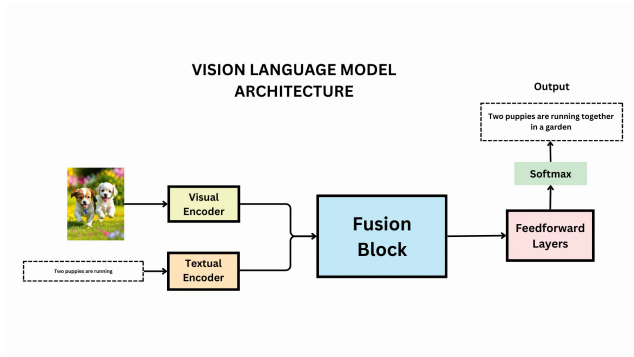
Improving Confidence Calibration in Vision-Language Models

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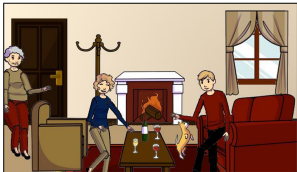


Vision-Language Models (VLMs)

- Jointly process images + text
- Components:
 - Vision Encoder
 - Language Model
 - Projection Block
- Applications:
 - Image captioning
 - Visual question answering (VQA)
 - Multimodal reasoning



Visual Question Answering (VQA)



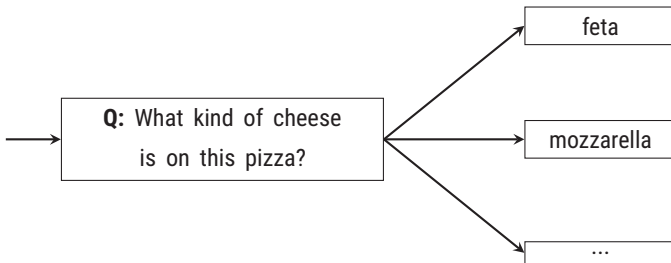
Q: How many glasses
are on the table?

3

Q: What is the
woman reaching for?

glass

What is Calibration?



Confidence should reflect the inherent uncertainty

Problem

- VLMs perform well on VQA tasks
- However, they are often **overconfident**:
confidence > accuracy

Existing Work

- Temperature scaling
- MC Dropout, Deep ensembles

Goal

- Improve **confidence calibration** in VLMs
- Reduce overconfident hallucinations
- Enable reliable abstention (*"I don't know"*)

Dataset: VQA v2

- Built on Microsoft Common Objects in Context
- Contains:
 - Images with complex real-world scenes
 - Object annotations and captions
 - Open-ended questions and 10 human answers per question



What kind of store is this?

bakery
bakery
pastry

art supplies
grocery
grocery

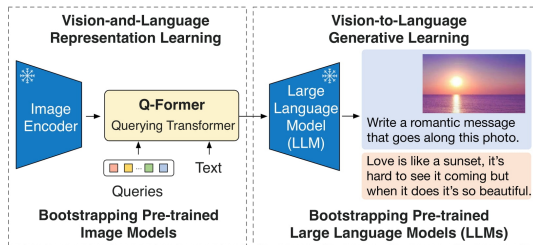
Is the display case as full as it could be?

no
no
no

no
yes
yes

Baseline Model (BLIP-2)

- Pretrained Vision-Language Model
- Components:
 - Image encoder
 - Language model
- Outputs: probability distribution over answers



Our Implementation

- Freeze backbone (BLIP-2)
- Apply Laplace approximation to output head
- Average outputs for confidence
- Model abstains when uncertain

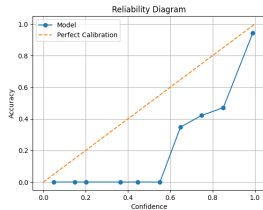
Baseline Methods

- Temperature scaling
- MC Dropout

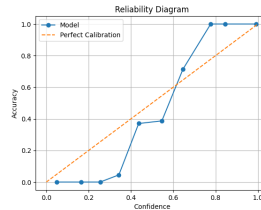
Evaluation Metrics

- Expected Calibration Error (ECE)
- Negative Log-Likelihood (NLL)
- Reliability diagrams

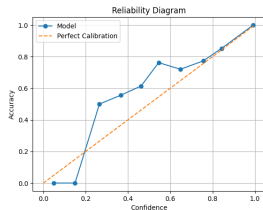
Results: Calibration Comparison



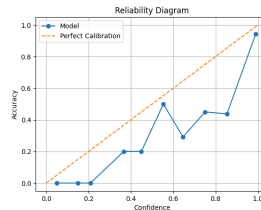
Baseline



Temp. Scaling



MC Dropout



Laplace Approx.

Method	Accuracy	ECE	NLL
Baseline	0.540	0.246	1.160
Temp Scaling	0.550	0.115	0.831
MC dropout	0.813	0.081	1.085
Laplace Approx	0.630	0.156	1.161

- MC Dropout achieves the lowest ECE and highest accuracy.
- Temperature scaling achieves the lowest NLL.
- The Laplace approximation shows limited improvement over baseline.

- VLMs like BLIP-2 can be overconfident on real VQA v2 questions.
- Post-hoc calibration (especially temperature scaling) significantly improves NLL and reduces ECE without sacrificing accuracy.
- MC Dropout achieves the best improvements in accuracy and ECE, indicating better uncertainty estimation, but comes with higher computational cost.
- In contrast, the Laplace approximation provides limited calibration benefit in this setup, suggesting that simple posterior approximations are insufficient for this task.

Thank you!

Questions?